Informatics in Medicine Unlocked 27 (2021) 100799

Contents lists available at ScienceDirect

Informatics in Medicine Unlocked

journal homepage: www.elsevier.com/locate/imu

Missing value imputation affects the performance of machine learning: A review and analysis of the literature (2010–2021)

Md. Kamrul Hasan [a,*], Md. Ashraful Alam [a], Shidhartho Roy [a], Aishwariya Dutta [b], Md. Tasnim Jawad [a], Sunanda Das [c]

[a] Department of Electrical and Electronic Engineering (EEE), Khulna University of Engineering & Technology (KUET), Khulna 9203, Bangladesh
[b] Department of Biomedical Engineering (BME), Khulna University of Engineering & Technology (KUET), Khulna 9203, Bangladesh
[c] Department of Computer Science and Engineering (CSE), Khulna University of Engineering & Technology (KUET), Khulna 9203, Bangladesh

ARTICLE INFO

Keywords:
Incomplete datasets
Imputation methods and evaluations
Machine learning classifiers and evaluations
PRISMA technique

ABSTRACT

Recently, numerous studies have been conducted on Missing Value Imputation (MVI), intending the primary solution scheme for the datasets containing one or more missing attribute's values. The incorporation of MVI reinforces the Machine Learning (ML) models' performance and necessitates a systematic review of MVI methodologies employed for different tasks and datasets. It will aid beginners as guidance towards composing an effective ML-based decision-making system in various fields of applications. This article aims to conduct a rigorous review and analysis of the state-of-the-art MVI methods in the literature published in the last decade. Altogether, 191 articles, published from 2010 to August 2021, are selected for review using the well-known Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) technique. We summarize those articles with relevant definitions, theories, and analyses to provide essential information for building a precise decision-making framework. In addition, the evaluation metrics employed for MVI methods and ML-based classification models are also discussed and explored. Remarkably, the trends for the MVI method and its evaluation are also scrutinized from the last twelve years' data. To come up with the conclusion, several ML-based pipelines, where the MVI schemes are incorporated for performance enhancement, are investigated and reviewed for many different datasets. In the end, informative observations and recommendations are addressed for future research directions and trends in related fields of interest.

1. Introduction

Data mining or big data analysis is accepted as an essential and challenging responsibility for numerous applications in everyday life, where a particular dataset for a favored problem is accumulated to conduct such an examination. A dataset is the heart of any learnable decision-making system for automated classification or regression tasks. However, the collected practical dataset typically contains an unusual proportion of missing values, irregular patterns (outliers), and redundancy with one or more attributes, which must be eliminated to build a generic trained model. The former missing values are encoded as NaNs, blanks, undefined, null, or any other place-holders in tradition [1]. There are many reasons for such missingness, which arise from various sources in the datasets. These include improper and mistaken data entries, data unavailability, data collection problems, missing sequence, incomplete features, files missing, incomplete information, etc. However, there are three types of mechanisms for the missing values in the datasets [2], namely, Missing Completely at Random (MCAR), Missing at Random (MAR), and Not Missing at Random (NMAR). Let us consider an attribute variable I_i connected to the ith sample as in Eq. (1).

$$I_i = \begin{cases} 1 & if\ ith\ sample\ responds \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

In response modeling, we suppose that $P\{I_i = 1\} = \theta_i$. The response probability θ_i may rely on a set of known P auxiliary information $x_i = (x_{1i}, \ldots, x_{ji}, \ldots, x_{pi})$ or samples' true label y_i. The first mechanism, MCAR, occurs when an instance's or observation's probability having a missing value for an attribute does not rely on either the known values or the lost data, which exists when the missing values are randomly distributed across all observations [2]. In MCAR, the missingness of information is unrelated to any study variable. Thus, the participants with wholly observed data are a random sample of all the participants

* Corresponding author.
E-mail addresses: m.k.hasan@eee.kuet.ac.bd (M.K. Hasan), alam1603001@stud.kuet.ac.bd (M.A. Alam), swapno15roy@gmail.com (S. Roy), aishwariyadutta16@gmail.com (A. Dutta), jawad1703006@stud.kuet.ac.bd (M.T. Jawad), sunanda@cse.kuet.ac.bd (S. Das).

https://doi.org/10.1016/j.imu.2021.100799
Received 19 September 2021; Received in revised form 5 November 2021; Accepted 15 November 2021
Available online 30 November 2021
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Nomenclature

AdB	AdaBoost
AHC	Agglomerative Hierarchical Clustering
AI	Artificial Intelligence
ANNMVI	Artificial Neural Network-based MVI
ARMVI	Association Rule-based MVI
BPCAMVI	Bayesian Principal Component Analysis-based MVI
C4.5MVI	C4.5-based MVI
CARTMVI	Classification and Regression Tree-based MVI
CbC	Constraint-based Clustering
CCMVI	Class Center-based MVI
CVRMSE	Coefficient of variation RMSE
DbC	Density-based Clustering
DbSCAN	Density-based Spatial Clustering of Applications with Noise
DT	Decision Tree
ECLAT	Equivalence Class Clustering and Bottom-up Lattice Traversal
ELMMVI	Extreme Learning Machine-based MVI
EMMVI	Expectation Maximization-based MVI
FCmC	Fuzzy C-means Clustering
FCMMVI	Fuzzy C-means-based MVI
FKMMVI	Fuzzy K-means-based MVI
FPG	Frequent-pattern Growth
GAMVI	Genetic Algorithm-based MVI
GbC	Grid-based Clustering
GKNNMVI	Grey KNN-based MVI
GMMbC	Gaussian Mixture Model-based Clustering
GMMMVI	Gaussian Mixture Model-based MVI
GRLSRMVI	Graph Regularized Local Self Representation-based MVI
HbC	Hierarchical-based Clustering
HCMVI	Hierarchical Clustering-based MVI
HDMVI	Hot Deck-based MVI
IDMVI	Information Decomposition-based MVI
IKNNMVI	Iterative KNN-based MVI
ILLSMVI	Iterative Local Least Square-based MVI
KmC	K-means Clustering
KMCMVI	K-means-based MVI
KMVI	Kernel-based MVI
KNN	K-nearest Neighbors
KNNMVI	K-nearest Neighbor-based MVI
LASSO	Least Absolute Shrinkage and Selection Operator
LDA	Linear Discriminant Analysis
LGBM	Light Gradient Boosting Machine
LLSMVI	Local Least Square-based MVI
LLSR	Linear Least Squares Regression
LR	Logistic Regression
LRMVI	Linear/ Logistic Regression-based MVI
LTSMVI	Least Trimmed Square-based MVI
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MAR	Missing at Random
MbC	Model-based Clustering
MCAR	Missing Completely at Random
MCIL	Mean Confidence Interval Length
MCM	Monte Carlo Methods
MCMCMVI	Markov Chain Monte Carlo-based MVI
MCMVI	Matrix Completion-based MVI
MIADMVI	Multiple Imputation by Absolute Deviation-based MVI
MICEMVI	Multiple Imputation by Chained Equations-based MVI
MISRTMVI	Multiple Imputation by Sequential Regression Tree-based MVI
ML	Machine Learning
MLPMVI	Multilayer Perceptron-based MVI
MMVI	Mean/ Median/ Mode-based MVI
MsC	Mean-shift Clustering
MSE	Mean Square Error
MVI	Missing Value Imputation
NB	Naive Bayes
NBMVI	Naïve Bayes-based MVI
NMAR	Not Missing at Random
NRMSE	Normalized RMSE
PbM	Partitioning-based Clustering
PCAMVI	Principal Component Analysis-based MVI
PCC	Pearson Correlation Coefficient
PCP	Percentage of Correct Predictions
PLR	Polynomial Regression
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
QRILCMVI	Quantile Regression Imputation of Left-censored Data-based MVI
RbC	Rule-based Classification
RF	Random Forest
RFMVI	Random Forest-based MVI
RMSE	Root Mean Square Error
RSTMVI	Rough Set Theory-based MVI
SBEMVI	Single Best Estimator-based MVI
SFSN	Similarity Fraction of the Same Neighbors
SGD	Stochastic Gradient Descent
SKNNMVI	Sequential KNN-based MVI
SMLR	Simple and Multiple Linear Regression
SOMMVI	Self-organizing Map-based MVI
SVDMVI	Singular Value Decomposition-based MVI
SVM	Support Vector Machine
SVR	Support Vector Regression
SVRMVI	Support Vector Regression-based MVI
TMVI	Tensor Completion-based MVI
WKNNMVI	Weighted KNN-based MVI
XGB	Extreme Gradient Boosting

assigned a particular intervention. Suppose the probability θ_i is independent of the study variable y_i. In that case, the missing values fall into the MCAR category, not depending on the dataset, observed or unobserved values. For example, if someone lost a schedule, it may be replaced by a schedule taken at random from the set of filled schedules. The second mechanism, MAR, happens when an instance's probability of having a missing attribute value may depend on that attribute [2]. In other words, the lost data on a partly missing variable (y_p) is related to some other wholly observed variables (x_w) in the analysis model but not to the values of y_p itself. Assume the response probability θ_i depends

on the observed values of y_i but not on the missing values of y_i. In that case, the missing data are called MAR, not relying on the unobserved values but the observed data. Say, for example, if consumption on food is missing in household surveys, but the household size is not missing, then we determine missing consumption on food by fitting a linear regression. The third remaining mechanism, NMAR, transpires when the probability of an instance having a missing value for an attribute may depend on the value of that attribute [2]. If the data do not characterize with MCAR or MAR, they fall into the category of MNAR. If θ_i relies on the missing variable y_i, then the nonresponse is called NMAR, where the missingness depends on the observed values of the dataset and the missing value. For example, respondents with significant income are less likely to report income, or respondents contaminated with Human Immunodeficiency Virus (HIV) respond less inclined to report their HIV status. However, a recent article in 2020 has reviewed those mechanisms in detail, which can be found in [2]. Therefore, the to-the-point scrutinization of these missingness mechanisms is excluded from this article's principal focus. Irrespective of the reasons, it is essential to handle missing data because any statistical results based on a dataset with non-random missing values could be biased. Missing data also introduces some degree of unavoidable ambiguity into a statistical analysis. Furthermore, many Machine Learning (ML) algorithms do not support data with missing values [3].

In this article, Missing Value Imputation (MVI) methods, along with their evaluations, are rigorously investigated and reviewed. The technical concepts, with respective pros and cons, of different MVI schemes and mathematical formulations of their evaluation metrics are systematically provided for assisting the researchers in getting those materials in a single article. Additionally, different ML models and their evaluation metrics, with sufficiently brief descriptions and formulations, are covered in this article. The recent trends of employed MVI methods, MVI's evaluation, ML models, and ML models' evaluation are also given in this review. This study also demonstrates that the appliance of suitable MVI methods with the ML models successfully enhances the performance of the decision-making actions, conferring the outcomes from many articles with different MVI and ML models for various datasets in different domains. However, the core contributions of this review article on the MVI methods will be supportive for the research community for the following contexts:

- Essential information concerning missing values, their missingness mechanisms, MVI methods with evaluation metrics, which contributes valuable knowledge to the researchers' community.
- Accumulates twelve years' (2010–August 2021) different MVI methods for missingness imputation, including recent trends of most commonly employed MVI techniques and their evaluation.
- Includes various ML models with their performance metrics from the selected papers, revealing the most commonly employed models' and evaluation metrics' trends.
- Provides sufficient grounds for proving that incorporating MVI methods can improve the ML models' performance, presenting the results from various articles concerning different approaches and domain datasets.
- Affords adequate descriptions of MVI methods and ML models with their respective evaluation and several recommendations for selecting them, which may aid beginner to expert-level researchers to develop a complete decision-making system for real-world designs in diverse application domains.

This review's literature exploration is conducted by following the well-known Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) strategy [4], as displayed in Fig. 1. PRISMA is an evidence-based minimum set of items aimed at helping authors report a wide array of systematic reviews and meta-analyses, primarily utilized to estimate the issues of a health care intervention, ensuring a transparent and complete research reporting. With the keyword, "missing value imputation", a total number of $n = 428$ articles are found for the MVI process from "Google scholar". The initially identified articles manifest some duplications in the literature ($n = 57$), which are eliminated afterward, providing $n = 371$ articles. Then, in the first screening round (first-level exclusion), the articles of non-English language ($n = 11$) and the review, database, or letter-type articles ($n = 12$) are excluded. Only the articles in the year range 2010 ~ August 2021 are retained in this round. This exclusion has outputted $n = 266$ articles in total (see in Fig. 1). Afterward, scrutinizing is carried out on all the picked articles from the previous steps, revealing that $n = 37$ articles have irrelevant research objectives, and $n = 21$ articles' methods are not matched. This second exclusion provides $n = 208$ articles (see in Fig. 1). Lastly, $n = 17$ articles are further excluded, as they neither explicitly present the MVI methods nor other employed strategies. Through the subsequent identification, screening, and eligibility testing, $n = 191$ articles are finally included to carry this aimed review process.

The remainders of the paper are organized as follows: Section 2 presents the overview of the MVI methods, technical details & categorization of various MVI methods with evaluation metrics, systematic review, and year-wise trends of those method utilizations. Section 3 accumulates different ML classifiers in the chosen 191 articles of this review, followed by their evaluation techniques. Section 4 presents the relative comparison of different discussed methods and provides extensive illustrations with the recent trends of utilization. This section also provides a comprehensive discussion with the analysis of the various computational pipeline employing ML classifiers and MVI methods on different datasets in many application domains, demonstrating the impact of MVI strategies on the decision-making results of the ML models. In addition, we provide several observations and recommendations in Section 4. Finally, the conclusions are outlined in Section 5.

2. Overview on MVI methods

This section presents a detailed overview and review of different MVI methods (see in Section 2.1) and their evaluation metrics (see in Section 2.2) comprehensively from the selected articles, applying the PRISMA approach (see in Fig. 1). To continue the review process, altogether 191 articles are selected (as described in the earlier section), which are displayed in Fig. 2, demonstrating the year-wise publication numbers from 2010 to August 2021 for the MVI techniques. This figure points out that the number of publications on the MVI method per year increases with a positive slope rapidly, especially after 2017. It is remarkable to mention that the number of publications in 2021 has declined because we have selected the articles until August 2021. Therefore, it necessitates a thorough review of those enormous articles' methodologies of MVIs. Many authors experimented with imputation achievement by showing missing value simulations over a taken dataset using several missingness rates. Again, some scholars studied minimal missing rates, e.g., less than 30%, while other authors concentrated on wide ranges of missing rates, such as 5 ~ 80% [2]. This article considers only the methodological perspectives of the MVIs in the decided 191 articles. However, Lin and Tsai [2] conferred and reviewed several articles that accounted different missing rates: less than 30%, between 30 ~ 50%, and greater than 50%. When the dataset includes a tiny quantity of missingness (for instance, less than 10% or 15%) of the entire dataset; the missing samples can be eliminated, called case deletion, without significantly influencing the ultimate decision-making outcome. On the other hand, careful attention is necessitated when the missing rate tops 15%. However, every domain dataset does not reflect such a thumb rule, especially where short quantities of missingness even carry the vital message, such as the records holding a very high amount of money spent by some consumers. Unlike the case deletion policy, MVI is the most regularly employed to deal with the fragmentary dataset enigma. In common, MVI is a means by which statistical or ML-based techniques are employed for restoring the missingness with newly filled values. The following Section 2.1 explains MVI procedures and describes the variety of its algorithms.

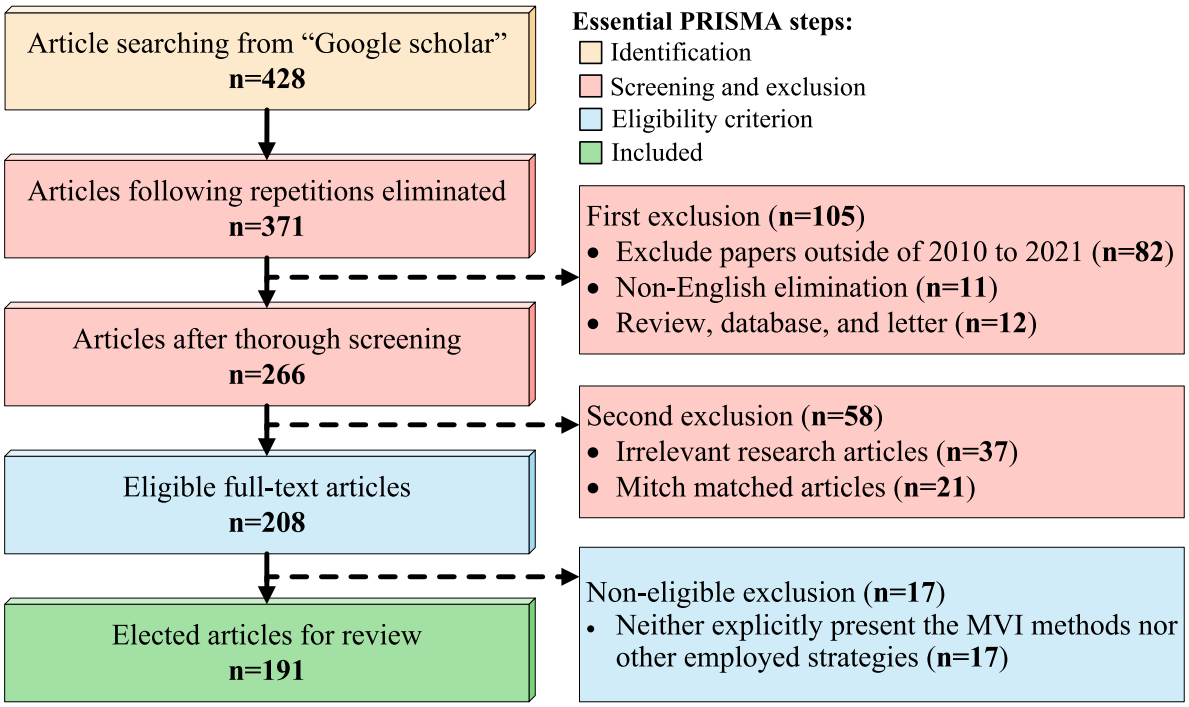


Fig. 1. The employed PRISMA method for article searching procedure, where we demonstrate the evidence for the article's inclusion and exclusion criteria.

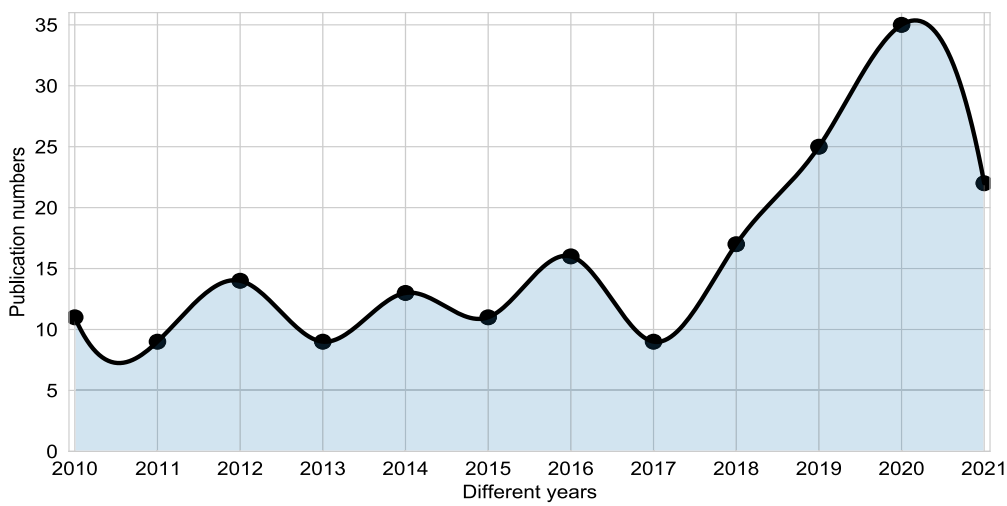


Fig. 2. Presentation of the number of per-year publications on the MVI method for the ML-based models. The number of publications in 2021 has been reduced due to the selection until August.

2.1. Imputation methods

Fig. 3 exhibits the experimental block diagram of the generic MVI method, where each incomplete dataset is divided into complete and missing sets. The former complete data is utilized to learn (train) parameters and estimate proper values by one of the many MVI methods (see in Fig. 4) to replace the missing values in the missing dataset. Finally, the most straightforward approach is to estimate the differences between the actual and imputed values to evaluate the imputation methods. The alternate technique includes utilizing the outputted complete dataset for classification or clustering and inspecting the achieved metrics. The details of the evaluation are explored in the next Section 2.2. However, plenty of methods for the MVIs [5–45] are found from the literature scrutinization, which are broadly classified into two categories [2]: statistical and ML-based MVI techniques, as exhibited in Fig. 4. Those two categories of MVI methods are further sub-categorized into different algorithm types (see in Fig. 4), which ease the comparative discussion. Those mentioned methods in Fig. 4 are broadly discussed in Table 1, also mentioning that the method is supervised or unsupervised. The essential concept of implementing those schemes is mentioned in Table 1 with respective pros or cons. The algorithmic details of those MVI methods for the implementation are also linked by providing their respective references or citations, as depicted in Table 1.

Some of those listed MVI methods in Table 1 are reported as the foundation for relevant experiments despite introducing a novel MVI method or being picked for the particular field of experiments, such as medical & financial decision-making systems, pattern classification, questionnaires, and industrial operation management. It necessitates mentioning that explaining the to-the-point theories of these MVI techniques is not the principal focus of this article. We, therefore, provide the citations in Table 1 for the corresponding MVI methods so that

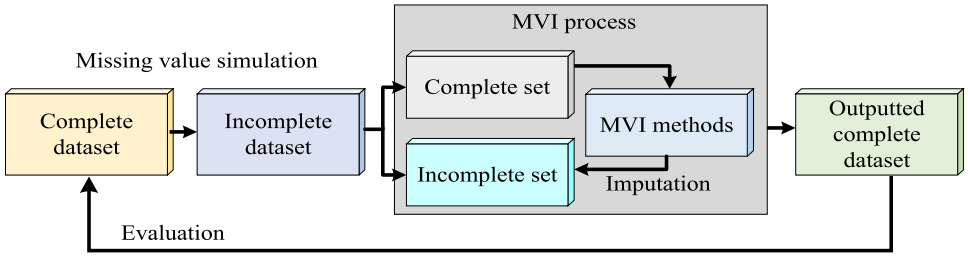


Fig. 3. The typical experimental configuration for MVI procedures to impute the missing values in any attributes [2].

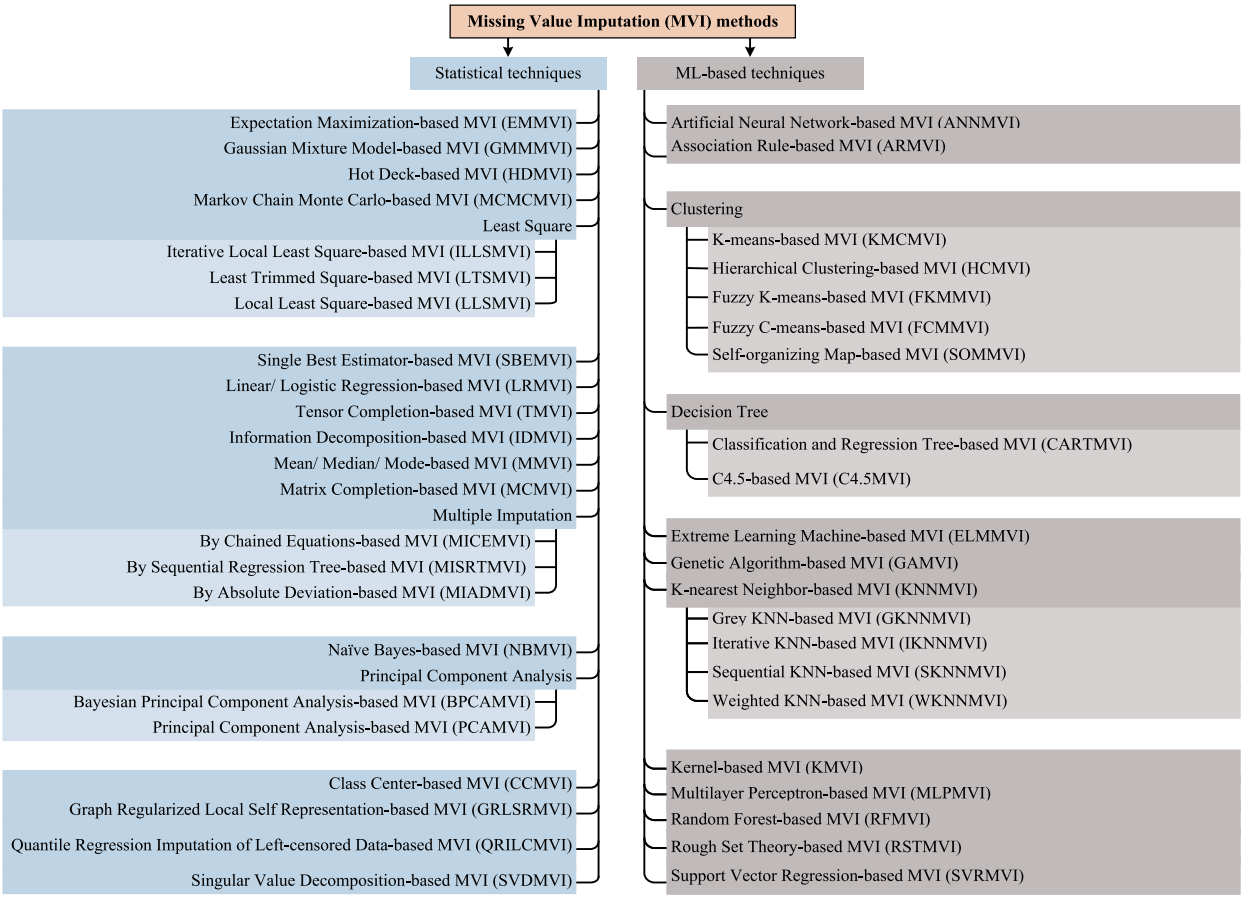


Fig. 4. The categorized tree exhibition of the commonly employed MVI methods, available in the literature.

the readers can explore them to attain detailed theoretical information. Most of the earlier surveys concentrated on explaining the fundamental ideas of the MVI systems, where the most widely practiced and MVI's effects on the ML-based decision-making systems are not clearly explained and reviewed. This article aims to provide a comprehensive survey of the MVI methods with other essential relevant studies. However, Table 2 records all of those statistical and ML-based MVI systems that are frequently employed in different published articles from 2010 to August 2021. This finding provides a primary guesstimation to the beginner to expert-level researchers to choose a suitable MVI algorithm for their decision-making pipeline. From Table 2, it is noteworthy that a few algorithms, such as EMMVI, HDMVI, LLSMVI, LRMVI, MMVI, MICEMVI, BPCAMVI, SVDMVI, ANNMVI, KMCMVI, FCMMVI, CARTMVI, KNNMVI, and RFMVI, are most regularly exercised in the last decade (2010–August 2021), compared to all other algorithms in Fig. 4.

Table 2 tells that statistical MVI, such as EMMVI and MMVI, and ML-based MVI, e.g., KNNMVI and RF, are massively employed imputation methods in the last decade (2010–August 2021). The E-step in EMMVI algorithm calculates the current parameter from the complete data, and then the M-step modernizes the parameter by the maximization of the likelihood function [171]. This process continues until it reaches the stopping criterion, and the missing values are determined from the updated likelihood function. In the MMVI method, an attribute's missing values are estimated from the mean (average), median (middle), or mode (most frequent) values of that attribute from the complete dataset. However, the normal distribution of any attribute has the same value of the mean, median, and mode, as experimentally proved in [172] for the MVI. In the KNNMVI policy, the missing values are filled utilizing the measured values from the k nearest observed values using a distance function, usually the Euclidean distance. The

Table 1
All different MVI methods of Fig. 4 with their respective citation, core imputation mechanism, and particular advantages or disadvantages. The symbols cross-mark (✗) and check-mark (✓) denote the unsupervised and supervised MVI methods, respectively.

Methods (Supervised?)	Technical concept	Remarks
Statistic-based MVI methods		
EMMVI [5] (✗)	Estimates parameters directly, maximizing the complete data log-likelihood function by iterating between the E-step and M-step.	Very slow convergence and not applicable when confidence intervals of estimated parameters are the primary goals.
GMMMVI [6] (✗)	Determines the missing value as data generation iteratively from the cluster, satisfying the condition of the log-likelihood function.	Provides unsatisfactory imputed values if the class likelihood and its cluster have a heavily shared or common region between the classes.
HDMVI [46] (✗)	Imputes missing values by adopting values from similar, but complete records of the same dataset, employing an object called the donor.	Although Gower's coefficient determines a donor, a single donor might be selected to provide multiple recipients, creating redundancy.
MCMCMVI [8] (✗)	Assigns the missing values from likelihood estimation based on Bayesian inference using the imputation and posterior steps.	Not straightforward to determine the convergence, but reliable to replace missing value either a low or high percentage of missing data.
ILLSMVI [9] (✓)	Predicts the missing data from the regressors, wherein each iteration, one variable is used as a response variable, and other variables serve as regressors.	Requires to be more careful in this setting to ensure that the separate regression models are consistent among them.
LTSMVI [10] (✓)	Starts from the KNN and is followed by iterative regressions in a transformed isometric log-ratio space.	Allows estimating the missing cells in the response with the multivariate information and outliers.
LLSMVI [11] (✓)	Least square-based method, applying many regressions using K neighbors taking K coherent genes instead of weighting or averaging K coherent genes.	Applicable to various biological and chemical data; robust and accurate missing value estimation method.
SBEMVI [12] (✗)	Simple substitution method, selecting neighboring proximity elements (min. distance from target place).	Neglecting variances, as it considers only neighboring elements, can produce erroneous imputation.
LRMVI [13] (✓)	Predicts the missing values from the regression coefficients of the fitted line or curve.	Massively dependent on the outliers present in the training samples.
TMVI [14] (✓)	Fills the missing elements, where the low rank of the tensor is often necessary to restrict the degree of freedom to avoid being an underdetermined problem.	More stable and accurate, especially when the missing sample entries are minimal, and it can propagate structure to fill more significant missing regions.
IDMVI [15] (✗)	Imputes values, combined with multidimensional data with the help of the fuzzy membership function.	Often needs KNN to find more information when the fuzzy membership cannot transfer information.
MMVI [25] (✗)	Replaces values with a mean/ median/ mode value of non-missing elements in the corresponding variable.	Reduces the variance of the imputed variables and does not preserve correlations between variables.
MCMVI [16] (✗)	Fills the values by solving a Matrix Completion (MC) problem through inexact augmented Lagrange multipliers, considering MC as an optimization problem.	It is a hurdle to find stopping criteria during optimization due to the oscillation of convergence-minima for some kinds of datasets.
MICEMVI [17] (✗)	Assigns values from the regression model iteratively until reaching the stopping criterion, where missing positions are initially filled with mean and variance.	It has a significant advantage over other methods in terms of flexibility but does not have the same theoretical grounds as other methods.
MISRTMVI [18] (✗)	Combined Multiple Imputations (MI) and single tree prediction, decomposing the prediction error to the expected value and variance of the MI prediction.	Raises ambiguity when the imputation number increases and the correlation among prediction models decreases.
MIADMVI [19] (✗)	Uses outer covariates to fill missing values iteratively based on the fitted conditional model until meeting the ending criterion.	Demands slightly more data analytic capacity and skills of programming.
NBMVI [20] (✓)	Fills the missing values according to the Bayes theorem-based probability estimation from real data.	Imputation error value depends on the size of complete data, instances with no missing values.
BPCAMVI [21] (✗)	Uses probabilistic latent variables applying PCA regression, Bayesian estimation, and EM algorithms.	Not computationally efficient as it applies those three algorithms simultaneously.
PCAMVI [22] (✗)	Generates imputed data from a PCA model and are evaluated by projecting them to the principal axis.	Only higher variances are considered to minimize the errors; some crucial information may be lost.
CCMVI [23] (✗)	Each class center and its distances are marked to identify a threshold used for missing value imputation.	Requires very less imputation time than the ML-based methods.
GRLSRMVI [24] (✗)	Rough-filled values are refined using local self-defined cost function by linearly mixing the nearby samples and graph regularization, forcing the nearby samples' deviation.	Provides a global and uniform framework for recovering missing values, incorporating local structure information of data.

(*continued on next page*)

missing information is applied as the testing stage, in which the entire and missing attributes express the input characteristics and yield class labels individually. In the RFMVI technique, various decision trees are assembled based on the bootstrapping method, and the ultimate forecasts are provided by averaged values or majority votes of each tree's prognostication. The imputation method of RFMVI is related to KNNMVI, where the inner and leaf nodes describe the inputted attributes and outputted class labels, respectively. The following section describes and studies the metrics for MVI method evaluations and comparisons.

2.2. Imputation evaluation

Once the missing values are imputed applying one of the MVI methods, as described in the earlier section, the next step is to assess that imputation results, which can be accomplished in two ways: direct evaluation and indirect evaluation by classification accuracy. Again,

Table 1 (*continued*).

Methods (Supervised?)	Technical concept	Remarks
ML-based MVI methods		
QRILCMVI [25] (✗)	Imputes missing values by randomly drawing from a truncated distribution estimated by quantile regression, carrying log-transformation to gain accuracy.	Designed explicitly for left-censored data.
SVDMVI [26] (✗)	Fills from a linear mixture of the k most eigen variables iteratively until a certain convergence threshold.	Has the significant advantages of handling datasets of various sizes, mixed types, and faster.
ANNMVI [27] (✓)	The missing values are filled from the prediction of the end-to-end auto-encoder.	Separates overlapped clusters of data but are heavily affected by highly skewed data.
ARMVI [28] (✗)	Association rules describe the dependency links among data entries in a dataset where all data, including the missing ones, are filled according to that rules.	Only database entries which exactly match the candidate patterns may contribute to the support of the candidate pattern.
KMCMVI [29] (✗)	Iteratively tries to partition the dataset into K distinct clusters based on the sum of the squared distance between the data points and the cluster's centroid.	Can group more or less linearly separable clusters and do not work well with the global cluster.
HCMVI [30] (✗)	Build a hierarchy of clusters based on a greedy method, which falls into two types: agglomerative (bottom-up) and divisive (top-down) approaches.	No apriori information about the number of clusters required and easy to implement, requiring more time complexity ($n^2 log(n)$) for n samples.
FKMMVI [31] (✗)	An approach for exploring the structure of a set of patterns, when the clusters are overlapping and are supposed to reflect the complex nature of data.	Makes the resulting algorithms less susceptible to getting stuck in a local minimum and provides a better tool when the clusters are not well separated.
FCMMVI [32] (✗)	Data assigns to every cluster with likelihood and repeats until the likelihood is maximized and converged.	Allows one datum to belong to two or more clusters and is frequently employed in pattern recognition.
SOMMVI [33] (✗)	Allows one datum to two or more clusters and is frequently employed in pattern recognition.	Relies on a predefined distance in feature space and does not behave so gently when using categorical data, even worse for mixed data.
CARTMVI [34] (✓)	Seeks predictors and cut points in the predictors that are used to split the sample. The cut points divide the sample into more homogeneous subsamples.	Can handle numerical data that are highly skewed or multi-modal and categorical predictors with either ordinal or nonordinal structure.
C4.5MVI [35] (✓)	C4.5 performs pruning by replacing a subtree in the decision tree with a single decision node that contains all the subtree decisions.	Unstable, meaning that a slight change in the data can lead to a significant shift in the structure of the optimal decision tree and requires more memory.
ELMMVI [36] (✓)	Consists of three layers: input, hidden, and output layers, based on empirical risk minimization theory; avoids multiple iterations and local minimization.	Faster than most existing neural network algorithms, as it is a single hidden layer feedforward network and yields promising performance.
GAMVI [37] (✗)	It is introduced to generate optimal sets of missing values, and information gain is used as the fitness function to measure individual imputation's performance.	Requires fewer data about the problem, but building an objective function and getting the suitable operators can be difficult and computationally expensive.
GKNNMVI [38] (✓)	Based on the grey coefficient, the valid attribute values derived from these nearest neighbors are used to predict the missing values.	It gives a normalized measure function, and mutual information-based feature relevance is embedded during estimating missing values.
IKNNMVI [39] (✓)	Imputes missing values by K-nearest neighbors via calculating the gray distance between the missing datum rather than traditional Euclidean distance methods.	Searches for the nearest neighbor instance with the same class label between the instance and the missing instance, reducing the time complexity with reduced errors.
SKNNMVI [47] (✓)	Fills by taking the mean of the NN genes in the complete set where Euclidean Distance is used as the distance metric to determine the NN genes.	Has improved accuracy and computational complexity over the conventional KNN-based and other maximum likelihood estimation-based methods.
WKNNMVI [40] (✓)	Weighted KNN is employed, where weight is the correlation between a missing dimension and available data values from other fields.	Performance depends on data quality and is sensitive to the scale of data, which is also computationally expensive.
KMVI [41] (✗)	Imputes missing values by drawing from the posterior predictive distribution of the missing data given the observed data.	Convergence is very slow and not applicable when confidence intervals of estimated parameters are the primary goals.
MLPMVI [42] (✓)	Construct an MLP model, feed the data without having any missing value and finally predict for the missing value instances.	Many MLP models have to be constructed when missing items appear in several attributes in a high-dimensional problem.
RFMVI [43] (✓)	A tree is grown from randomly selecting training samples, as much as possible without pruning, where the top trees' features are more important than end nodes.	Found to be biased while dealing with categorical variables. Many trees can make the algorithm too slow and ineffective for real-time predictions.
RSTMVI [44] (✓)	The rough set rule induction method enables obtaining association rules of missing data patterns based on approximations, dependencies, and decision rules.	Does not need any preliminary or additional information, such as a probability distribution of the given data samples.
SVRMVI [45] (✓)	Estimates parameters directly, minimizing the distance (max. distance and Hausdorff distance) between the actual and prediction, using a ϵ-SVR model.	Suited for only interval data and reliant on the outliers and classes overlapping. It also underperforms for more features with fewer training samples.

there are mainly two kinds of values in the attributes: discrete or categorical and continuous. The former categorical value imputations are generally directly evaluated by estimating the Percentage of Correct Predictions (PCP) [2,18,120] and Cross-Entropy (CE) [173] (see their mathematical definitions in Table 3). The later continuous value imputations are directly assessed by computing the Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE) [2], Root Mean Square Error (RMSE) [2], RMSE Improvement (RMSEI) [58], Similarity

Table 2
The statistical and ML-based techniques, applied in the literature from 2010 to 2021. The publications in 2021 have been selected until August.

Methods	Related publications
	Statistic-based MVI methods
EMMVI	Aussem and de Morais [48], Ghannad-Rezaie et al. [49], Hron et al. [10], Jerez et al. [42], Polikar et al. [50], Di Nuovo [51], García et al. [52], Luengo et al. [53], Stekhoven and Bühlmann [54], Zhu et al. [55], Ding and Ross [56], Doquire and Verleysen [57], Kang [58], Yozgatligil et al. [59], Aisha et al. [60], Friedman [61], Zhang et al. [62], Li and Parker [63], Zainuri et al. [64], De Souto et al. [65], Ding and Song [66], Folino and Pisani [67], Ghorbani and Desmarais [68], Łuczyńska et al. [69], Aydin and ŞENOĞLU [70], Le et al. [71], Mazzutti et al. [72], Rafsunjani et al. [73], Wang et al. [74], Malan et al. [75], Rumaling et al. [76], Selvi et al. [77], Thulare et al. [78], Cook [79].
GMMMVI	García-Laencina et al. [80], Kang [58], Liu et al. [81], Mazzutti et al. [72], Chiu et al. [82], Mesquita et al. [83], Zhang et al. [84].
HDMVI	Jerez et al. [42], Ghorbani and Desmarais [68], De Waal et al. [85], Vazifehdan et al. [86], Song et al. [87], Christopher et al. [88], Santos and Conde [89], Xu et al. [19], Cook [79], Silva-Ramírez and Cabrera-Sánchez [90].
MCMCMVI	Ding and Ross [56], Yozgatligil et al. [59], Saha et al. [91], Suhaimi et al. [8], Anindita et al. [92], Gómez-Rubio and Rue [93].
ILLSMVI	Hron et al. [10], Cheng et al. [94], Chiu et al. [95], Rahman and Islam [96], Pati and Das [47].
LTSMVI	Hron et al. [10].
LLSMVI	Celton et al. [97], Yu et al. [98], Liu et al. [99], Oh et al. [100], Luengo et al. [53], Cheng et al. [94], Chiu et al. [95], Rahman and Islam [96], Rao et al. [101], Wang et al. [102], Al-Janabi and Alkaim [103], Wang et al. [104].
SBEMVI	Jahan et al. [12].
LRMVI	Jerez et al. [42], Di Nuovo [51], Dumedah and Coulibaly [105], Silva-Ramírez et al. [7], Zhu et al. [55], Eirola et al. [106], De Souto et al. [65], Silva-Ramírez et al. [107], Shao et al. [108], Jahan et al. [12], Cook [79].
TMVI	Liu et al. [109], Ng et al. [110], Bengua et al. [111], Long et al. [112], Xie et al. [113].
IDMVI	Liu et al. [114].
MMVI	Hron et al. [10], Jerez et al. [42], Polikar et al. [50], Silva-Ramírez et al. [7], Ding and Ross [56], Luengo et al. [53], Senapti et al. [115], Rao et al. [101], Kang [58], Liao et al. [116], Tian et al. [117], Gromski et al. [118], Zainuri et al. [64], Silva-Ramírez et al. [107], Armitage et al. [119], De Souto et al. [65], Valdiviezo and Van Aelst [18], Folino and Pisani [67], Nishanth and Ravi [120], Huang et al. [121], Chen et al. [122], Kiasari et al. [123], Mesquita et al. [124], Pati and Das [47], Xia et al. [125], Le et al. [71], Maniruzzaman et al. [126], Wei et al. [25], Rafsunjani et al. [73], Christopher et al. [88], Selvi et al. [77], Xu et al. [19], Cook [79], Rani et al. [127].
MCMVI	Thulare et al. [78].
MICEMVI	Jerez et al. [42], Janssen et al. [128], Burgette and Reiter [129], Ding and Ross [56], Stekhoven and Bühlmann [54], Liao et al. [116], Valdiviezo and Van Aelst [18], Garciarena and Santana [130], Rafsunjani et al. [73], Rani et al. [127].
MISRTMVI	Valdiviezo and Van Aelst [18].
MIADMVI	Xu et al. [19].
NBMVI	Subasi et al. [131], Leung and Leung [132], Aisha et al. [60], Nishanth and Ravi [120].
BPCAMVI	Celton et al. [97], Liu et al. [99], Yu et al. [98], Oh et al. [100], Cheng et al. [94], Luengo et al. [53], Chiu et al. [95], Rahman and Islam [96], Rao et al. [101], Pati and Das [47], Xia et al. [125].
PCAMVI	Zuccolotto [133], Van Ginkel and Kroonenberg [134], Chen et al. [122], Saha et al. [91], Malan et al. [75].
CCMVI	Tsai et al. [23].
GRLSRMVI	Chen et al. [135].
QRILCMVI	Wei et al. [25].
SVDMVI	Liu et al. [99], Yu et al. [98], Oh et al. [100], Luengo et al. [53], Rao et al. [101], Zainuri et al. [64], Saha et al. [91], Paul et al. [136], Wei et al. [25], Rafsunjani et al. [73]
	ML-based MVI methods
ANNMVI	Canchala-Nastar et al. [137], Choudhury and Pal [27], Shao et al. [108], Nishanth and Ravi [120], Gautam and Ravi [138], Aydilek and Arslan [139].
ARMVI	Vougas et al. [140], Chhabra et al. [141], Li et al. [142].
KMCMVI	Raja and Thangavel [143], Chhabra et al. [141], Migdady and Al-Talib [31], Kang [58], Eirola et al. [106], Luengo et al. [53], Somasundaram and Nedunchezhian [144].
HCMVI	Celton et al. [97].
FKMMVI	Pan et al. [145], Luengo et al. [53].
FCMMVI	Raja and Thangavel [143], Zhang et al. [146], Tian et al. [117], Aydilek and Arslan [147], Somasundaram and Nedunchezhian [144], Di Nuovo [51], Li et al. [148].
SOMMVI	Singh et al. [33], Folguera et al. [149], Kang and Yusof [150], Jerez et al. [42], Merlin et al. [151].
CARTMVI	Purwar and Singh [1], Doove et al. [152], Hapfelmeier et al. [153], Ghannad-Rezaie et al. [49], Ding and Simonoff [154], Burgette and Reiter [129].
C4.5MVI	Nishanth and Ravi [120], Ding and Simonoff [154].
ELMMVI	Christo et al. [155], Zhang et al. [156], Lu and Mei [157], Shao et al. [108].
GAMVI	Lin and Tsai [2], Mishra et al. [158], García et al. [52].

(*continued on next page*)

Table 2 (*continued*).

Methods	Related publications
KNNMVI	Dong et al. [159], Liu et al. [160], Lee and Styczynski [161], Paul et al. [136], Nishanth and Ravi [120], Tsai and Chang [162],Armitage et al. [119], De Souto et al. [65], Valdiviezo and Van Aelst [18], Pan et al. [145], Tian et al. [117], Rao et al. [101], Eirola et al. [163], Chiu et al. [95], Stekhoven and Bühlmann [54], Ding and Ross [56], Doquire and Verleysen [57], Kang [58], Luengo et al. [53], Hron et al. [10], Celton et al. [97], Hron et al. [10], Jerez et al. [42].
GKNNMVI	Zhu et al. [55], Zhang [39].
IKNNMVI	Chiu et al. [95].
SKNNMVI	Pati and Das [47], Chiu et al. [95].
WKNNMVI	Shao et al. [108], Luengo et al. [53].
KMVI	Zhu et al. [55].
MLPMVI	Jung et al. [164], Choudhury and Pal [27], García-Laencina et al. [80], Silva-Ramírez et al. [7], Gautam and Ravi [138], Jerez et al. [42]
RFMVI	Kokla et al. [43], Khan and Zubair [165], Wei et al. [25], Xia et al. [125], Xia et al. [125], Nishanth and Ravi [120], Purwar and Singh [1], Valdiviezo and Van Aelst [18], Kapelner and Bleich [166], Shah et al. [167], Doove et al. [152], Hapfelmeier and Ulm [168].
RSTMVI	Sujatha et al. [169], Clark et al. [170].
SVRMVI	Aydilek and Arslan [139], Luengo et al. [53].

Angle (SA) [174], Similarity Length (SL) [174], Similarity Fraction of the Same Neighbors (SFSN) [174], Pearson Correlation Coefficient (PCC) [175], R^2 [7,96,119,176,177], and Mean Confidence Interval Length (MCIL) [178] (see their mathematical definitions in Table 3). Again, Gu et al. [174] estimated the modified RMSE (mRMSE) value from both the categorical and continuous attributes as $mRMSE = \sqrt{{RMSE_0}^2 + {RMSE_1}^2}$, where $RMSE_0$ and $RMSE_1$ are RMSEs of categorical and continuous variables, respectively and are defined in Eq. (2).

$$RMSE_0 = \sqrt{\frac{1}{|M_0|}\sum_{(i,j)\in M_0} 1_{\{\hat{x_{ij}} \neq x_{ij}\}}},$$
$$RMSE_1 = \sqrt{\frac{1}{|M_1|}\sum_{(i,j)\in M_0} (\hat{x_{ij}} - x_{ij})^2}, \tag{2}$$

where the known values $\{x_{ij} : (i,j) \in M_0 \cup M_1\}$ in X are the ground truth in evaluation. In several articles, the authors also modified the RMSE metrics, such as Normalized RMSE (NRMSE) [98,100,175,179–184] and Coefficient of variation RMSE (CVRMSE) [59]. The former NRMSE is achieved by normalizing the calculated RMSE value, whereas the latter CVRMSE is acquired by dividing the RMSE by the mean of actual values. Mean Square Error (MSE) [7,185–190] and Normalized MSE (NMSE) Sorjamaa et al. [190] are also determined and applied in several articles, which also can be calculated from the RMSE value. The MSE is computed as $MSE = RMSE^2$, and NMSE is the normalized form of the MSE. Sometimes, Total Computation Time (TCT) is also utilized as a metric to evaluate the MVI methods for finding the time required by the algorithm to impute the missing values, either categorical or continuous. The required TCT depends on the dataset size and the missing rate. However, among the reviewed articles, a few papers considered the TCT for their MVI evaluation strategy; these are [18,54,63,91,117,123,135,167,191–195].

The alternative indirect evaluation strategy examines the classification metrics (see details in Section 3.2) of some elected classifiers trained by the outputted complete dataset with imputed values. Unlike the direct evaluation approach, the imputed dataset after the imputation is utilized for training some particular classification models (see detailed models in Section 3.1). The classifier with better outcomes concerning the same metric indicates the better imputation quality of its inputted dataset, consequently outputting better imputation methods. However, Table 4 exhibits the categorization of those selected 191 articles depending on their direct and indirect MVI evaluation, which reveals that direct methods are more engaged in contrasting indirect assessment.

It is unquestionably noticed from Table 3 that the direct evaluation strategies have been hugely employed since the last decade (2010–August 2021) for the missingness imputation. Additionally, the careful one-by-one scrutinization of all the selected 191 articles exposes that the proportion of works utilizing both the evaluation policies is significantly shorter, corresponding to the direct and indirect evaluation strategies. The articles related to the employment of both the MVI method evaluation systems are [86,95,180,186,188,197,199,209,211,228].

3. ML models and their evaluation

This section briefly introduces various ML models in Section 3.1 and the evaluation metrics for the ML models in Section 3.2. The survey and the employment of those two items in selected articles from 2010 to August 2021 are also presented in this section.

3.1. ML models

Artificial Intelligence (AI), particularly ML, has been overgrown in recent years in data analysis and computing that typically allows the applications to function intelligently. ML usually provides systems with the ability to learn and enhance from experience without being specifically programmed automatically and is generally referred to as the most popular latest technology in the fourth industrial revolution. Additionally, it has been applied to the digital world, such as the internet of things, cybersecurity, mobile data, business, social media, health data, etc [287]. The ML algorithms can be categorized into four primary types: supervised, semi-supervised, unsupervised, and reinforcement learning.

Supervised learning, also known as the task-driven approach, is typically the responsibility of the ML model to learn a function that maps an input to an output based on sample input–output pairs [288]. It utilizes labeled training data and a gathering of training instances to understand a function. The most common supervised tasks are classification and regression. The former task divides the data into the classes, and the latter regression task fits the data. **Unsupervised learning**, a data-driven process, investigates unlabeled datasets without the necessity for intervention, which is extensively applied for obtaining generative features, recognizing significant trends and compositions, groupings in results, and experimental ideas [288]. The most popular unsupervised learning assignments are clustering, density estimation, feature learning, dimensionality reduction, finding association rules, anomaly detection, etc. **Semi-supervised learning** can be described as a hybridization of supervised and unsupervised techniques mentioned

Table 3
List of the commonly-applied evaluation metrics for imputation of the missing values.

Metrics	Definitions	Comments
PCP	$PCP = \frac{CP}{TP} \times 100$	CP and TP denote correct and total prediction numbers. PCT measures the percentage of correctly or incorrectly imputed values and is used for discrete value imputation.
CE	$H(P,Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)}$	P and Q represent the joint distributions of actual and predicted values. It computes the cross-entropy from the actual distribution to the distribution predicted from the models and applies for discrete value imputation.
MAE	$MAE = \frac{1}{n} \sum_{i=1}^{n} \lvert x_i - \hat{x}_i \rvert$	x_i and $\hat{x}_i$ indicate actual and imputed values for n samples. MAE estimates the mean error of the predicted and actual values and evaluates continuous value imputation.
MAPE	$MAPE = \frac{100}{n} \sum_{i=1}^{n} \lvert \frac{x_i - \overline{x_i}}{x_i} \rvert$	x_i and $\hat{x}_i$ indicate actual and imputed values for n samples. It measures the mean absolute percentage error that has been imputed in the missing positions for continuous attributes.
RMSE	$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (x_i - \overline{x_i})^2}$	x_i and $\hat{x}_i$ indicate actual and imputed values for n samples. It measures the root mean square error for the predicted continuous variables concerning the actual variables.
RMSEI	$RMSEI = 1 - \frac{RMSE_1}{RMSE_0}$	$RMSE_1$ and $RMSE_0$ indicate RMSE with/ without imputation. It calculates the percentage of RMSE from the RMSE values with/ without imputation for the continuous attributes.
SA	$SA = \sum_{i=1}^{3} \arccos(x_i - \hat{x}_i)$	x_i and $\hat{x}_i$ indicate actual and imputed values for the first three principal components. It estimates the sum of angles of the first three principal components calculated for the actual and imputed values and is employed for continuous variables.
SL	$SL = \sum_{i=1}^{3} \sqrt{\sum_{j=1}^{3} \lvert\lvert V(x_i - \hat{x}_i) \rvert\rvert}$	$\lvert\lvert V(x_i - \hat{x}_i) \rvert\rvert$ represents the length of projections of x_i and $\hat{x}_i$. It measures the sum of the lengths of projections for the actual and imputed values and works for continuous variables.
SFSN	$SFSN = \frac{1}{n} \sum_{i=1}^{n} \frac{\lvert M_i \cap \hat{M}_i \rvert}{\lvert M_i \cup \hat{M}_i \rvert}$	M_i and $\hat{M}_i$ are the k nearest neighbors' coordinates of x_i and $\hat{x}_i$, respectively. It calculates the mean fraction of the same neighbors from the k nearest neighbors between actual and predicted continuous values.
PCC	$PCC = \frac{\sum(x_i - \bar{x})(\hat{x}_i - \bar{\hat{x}})}{\sqrt{\sum(x_i - \bar{x})^2 (\sum \hat{x}_i - \bar{\hat{x}})^2}}$	x_i and $\hat{x}_i$ indicate actual and imputed values for n samples. $\bar{x}$ and $\bar{\hat{x}}$ are the mean values of x and $\hat{x}$. It estimates the linear correlation between data with/ without imputation of continuous variables.
R^2	$R^2 = 1 - \frac{\sum_{i=1}^{n}(x - \hat{x})^2}{\sum_{i=1}^{n}(x - \bar{x})^2}$	x_i and $\hat{x}_i$ indicate actual and imputed values for n samples and $\bar{x}$ is the mean value of x. It is a statistical measure that indicates how well the predicted values are close to the real data values.
MCIL	Confidence interval refers to the probability that a population parameter will fall between a set of values for a certain proportion of times.	
TCT	Elapsed time for training and testing the models for the imputation.	

above, as it works on both labeled and unlabeled data. Hence, it befalls between learning without supervision and learning with supervision. In real-world scenarios, labeled data could be limited in sparse settings, and unlabeled data are numerous, where semi-supervised training is beneficial. The terminal goal of a semi-supervised learning model is to present a more favorable outcome for prognostication than that created by applying the labeled data solely from the model. Some employment fields where semi-supervised learning is utilized incorporate machine translation, fraud detection, labeling data, and text classification [287,288]. **Reinforcement learning** is an ML algorithm that facilitates software tools and machines to automatically assess the optimal performance in a unique connection for increasing productivity, i.e., an environment-driven procedure. Such type of training is based on award or punishment, and its final purpose is to utilize perspicacity received from environmental activists to improve the reward or minimize the uncertainty. It is a mighty tool for training AI models that can support expanded automation or optimize the operational effectiveness of well-bred methods, such as robotics, autonomous driving tasks, manufacturing, and supply chain logistics. However, it is not superior to utilize it to determine the fundamental or straightforward difficulties [288].

Those discussions conclude that the ML models can be arranged into five groups: *Classification*, *Regression*, *Clustering*, *Association rule learning*, and *Reinforcement learning*. Fig. 5 displays the available ML model in the elected 191 articles for this review, where we exhibit all the five groups mentioned above for ease of presentation. The first category, *classification*, is a learning method to a predictive modeling problem, where a class label is predicted for a given example [287]. The classification model includes eleven different strategies: NB [289], DT [290], SVM [291], SGD [292], LGBM [293], XGB [294], RbC [295], KNN [296], RF [297], LR [298], and LDA [299]. The second group, *regression analysis*, predicts a continuous (y) result variable based on the value of one or more (x) predictor variables which includes five different models: SMLR [287], LASSO [300], PLR [301], SVR [302], and LLSR [303]. The third family, *clustering*, is an unsupervised ML technique for identifying and grouping related data points in large datasets without concern for the specific outcome which covers twelve different approaches: DbSCAN [304], GbC [305], HbC [306], DbC [307], PbM [308], CbC [309], MsC [310], GMMbC [311], AHC [312], KmC [313], MbC [314], and FCmC [315]. The fourth category, *association rule learning*, is a rule-based ML approach to discover interesting relationships, "IF-THEN" statements, in large datasets between variables. This includes six different models, such as AIS [316], SETM [317], Apriori [318], FPG [319], ABC-RuleMiner [320], and ECLAT [321]. The fifth group, *reinforcement learning*, is an ML technique that allows an agent to learn by trial and error in an interactive environment using input from its actions and experiences [287] which embraces three methods: MCM [322], QL [323], and DQL [324].

After carefully scrutinizing those chosen papers, the top most 15 operating ML models are presented in Fig. 6. This figure explicates that KNN is the topmost classifying method in those 191 articles, which is three times more than the second-highest RF model. It is also observed from Fig. 6 that SVM, BPCA, and DT are applied in an almost identical number of articles, occupying third, fourth, and fifth-most employed ML models, respectively. Quick calculation time, simple algorithm to interpret, versatile usefulness for regression and classification, and evolving with new data are the conceivable reasons for the massive employment of the KNN model in the literature.

Table 4
Year-wise accumulation of different 191 articles for indirect and direct evaluation approaches to assess the MVI methods.

Year	Indirect evaluation	Direct evaluation
2021	Ribeiro and Freitas [196], Rashid and Gupta [197], Jin et al. [180], Khan et al. [198], Kim et al. [188], Li et al. [199]	Aidos and Tomás [200], Bansal et al. [191], Hamid et al. [201], Wang and Liu [202], Song and Yu [175], Siswantining et al. [203], Rashid and Gupta [197], Rashid et al. [192], Zhu et al. [182], Michel et al. [204], Dong et al. [205], Gad et al. [189], Gong et al. [206], Jin et al. [180], Kim et al. [188], Lai et al. [207], Li and Chiou [208], Li et al. [199]
2020	Rosado-Galindo and Dávila-Padilla [209], Son and Kim [210], Lavanya and Mallika [211], Liu et al. [212], Payrovnaziri et al. [213]	Raja and Thangavel [143], Rosado-Galindo and Dávila-Padilla [209], Rumaling et al. [76], Ryu et al. [214], Sadhu et al. [176], Ye et al. [193], Zhao and Udell [215], Zhao et al. [216], Qiu et al. [217], Manna and Pati [183], M Mostafa et al. [177], Liu et al. [218], Lai et al. [194], Landgrebe et al. [219], Lavanya and Mallika [211], Di Lena et al. [220], Lenz et al. [184], Karmitsa et al. [221], Agbo et al. [222], Cho et al. [223], Gong et al. [224], Gupta and Gupta [225], Ispirova et al. [226]
2019	Berg et al. [227], Nekouie and Moattar [228], Rafsunjani et al. [73], Ribeiro and Freitas [229], Saha et al. [230], Vazifehdan et al. [86]	Gupta et al. [231], Arslan et al. [232], Biessmann et al. [187], Christopher et al. [88], Delaporte et al. [233], Koshechkin et al. [195], Lee et al. [234], Liu et al. [160], Nekouie and Moattar [228], Saha et al. [230], Sefidian and Daneshpour [235], Vazifehdan et al. [86]
2018	Chhabra et al. [236], Huang et al. [237]	Jin et al. [238], Abidin et al. [239], Ae Lee and Gill [240], Bischof et al. [241], Chen et al. [135], Demirhan and Renwick [242], Zhang et al. [243], Li et al. [244], Ramezani et al. [245], Wang et al. [102], Wei et al. [25]
2017	Zeng et al. [186], Garciarena and Santana [130], Ghorbani and Desmarais [68], Kiasari et al. [123], Xia et al. [125]	Armina et al. [246], Anindita et al. [92], Faisal and Tutz [247], Karim et al. [248], Shao et al. [108], Susanti and Azizah [249], Wu and Jhou [250], Zeng et al. [186]
2016	Al Khaldy and Kambhampati [251], Darryl and Rahman [252], Madhu and Nagachandrika [253], Raja and Thangavel [254], Tsai and Chang [162]	Deb and Liew [255], Di Guida et al. [256], Keerin et al. [257], Kumar et al. [258], Ma and Zhong [259], Rahman and Islam [260], Yang et al. [261]
2015	Purwar and Singh [1], De Souto et al. [65], Pan et al. [145]	Armitage et al. [119], Li et al. [262], Nookhong and Kaewrattanapat [263], Rachmawan and Barakbah [264], Stekhoven and Bühlmann [54], Yan et al. [6]
2014	Clark et al. [170], Li and Parker [63]	Asian et al. [265], Chiou et al. [266], Insuwan et al. [267], Li et al. [268], Liao et al. [116], Rahman and Islam [5], Wellenzohn et al. [269]
2013	Chiu et al. [95], Rahman and Davis [270], García-Laencina et al. [80], Leung and Leung [132], Kang [58]	Cheng et al. [271], Chiu et al. [95], Kang [58], Keerin et al. [272], Rahman and Islam [96], Wang et al. [273], Yozgatligil et al. [59]
2012	Miyakoshi and Kato [274], Rahman and Davis [275], Senapti et al. [115], Zhang et al. [276], Zhu et al. [55], Doquire and Verleysen [57], Luengo et al. [53], Nishanth et al. [277]	Dorri et al. [181], Friedman et al. [278], Gajawada and Toshniwal [279], Keerin et al. [280], Stekhoven and Bühlmann [54], Thirukumaran and Sumathi [281]
2011	Di Nuovo [51], Oh et al. [100]	Hsu et al. [282], Kim et al. [283], Oh et al. [100], Rahman and Islam [284], Silva-Ramírez et al. [7]
2010	Ghannad-Rezaie et al. [49], Polikar et al. [50]	Celton et al. [97], Ching et al. [11], Li et al. [285], Merlin et al. [151], Patil et al. [29], Ryan et al. [286], Sorjamaa et al. [185], Sun et al. [179], Yu et al. [98]

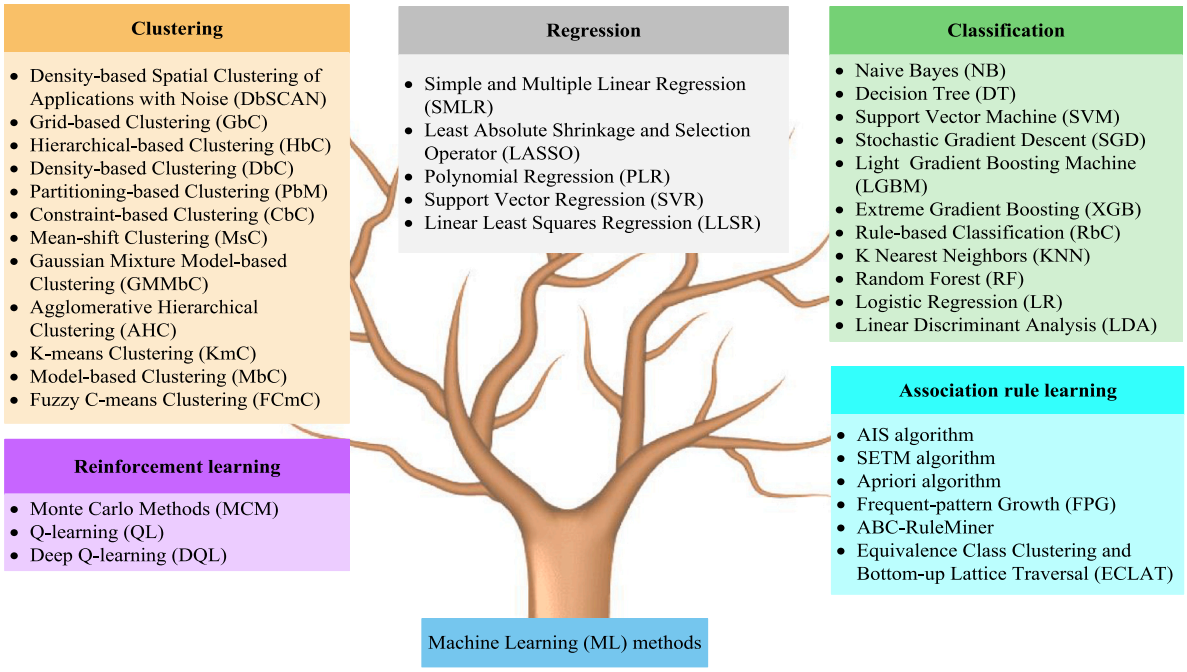


Fig. 5. Tree diagram of different ML models, where we group them into five categories depending on their similarity.

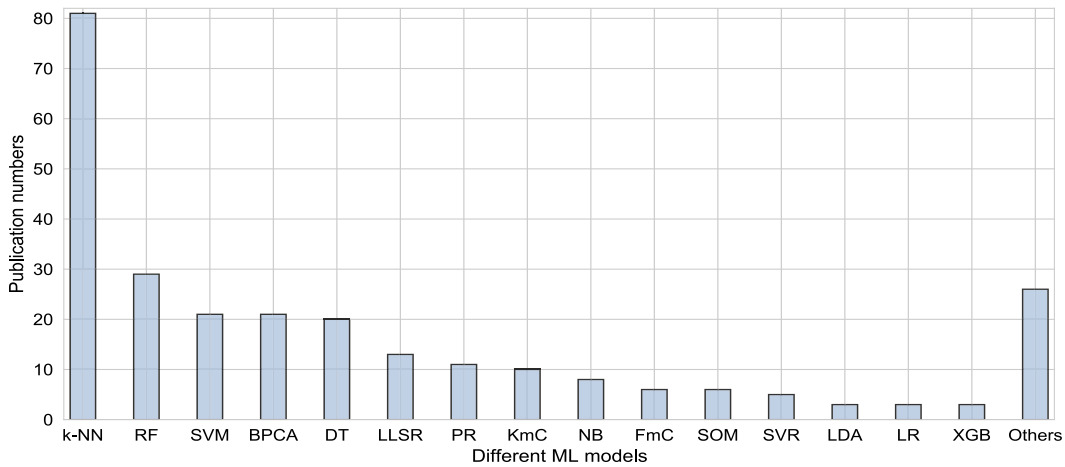


Fig. 6. Illustration of the number of articles on fifteen different ML models in the selected 191 articles, where the deep learning and multi-layer perceptron are included in the other category.

Table 5
Confusion matrix to define TP, TN, FN, and FP for evaluating a classifier.

		Predicted condition	
	Total population = P + N	Predicted condition positive (PP)	Predicted condition negative (PN)
Actual condition	Actual condition positive (P)	True Positive (TP) (Correct detection)	False Negative (FN) (Type-II error, underestimation)
	Actual condition negative (N)	False Positive (FP) (Type-I error, overestimation)	True Negative (TN) (Correct rejection)

3.2. ML classifier evaluation

The achievement of classifiers is usually recorded in a confusion matrix, as demonstrated in Table 5, which holds statistics about actual and predicted classes and sets the necessary associations needed to know accuracy computations for a particular classifier. In that matrix (see in Table 5), each column implies anticipated occurrences, while each row describes actual examples. The correct anticipations, such as True Positive (TP) and True Negative (TN), respectively indicate that the disease has successfully been identified when a condition is present. In contrast, the disease has successfully been rejected when an infection is not present. A False Negative (FN) or false-negative error is a test outcome that wrongly indicates that a condition does not hold. In contrast, a False Positive (FP) or false-positive error is a decision that means a given state exists when it does not. To evaluate different ML-based classifiers for the binary categorization, the most commonly applied metrics are True-positive Rate (or Sensitivity, or Recall) (TPR), True-negative Rate (or Specificity, or Selectivity) (TNR), False-positive Rate (or False alarm, or Fall-out) (FPR), False-negative Rate (or Miss rate) (FNR), Positive Predictive Value (or Precision) (PPV), Negative Predictive Value (NPV), False Discovery Rate (FDR), False Omission Rate (FOR), Accuracy, F1 score, Receiver Operating Characteristic (ROC) curve, Area under the ROC curve (AUC), Precision–recall (PR) curve, and Area under the PR curve (AP) (see their mathematical details in Table 6).

The multi-class evaluation metrics are the extensions of the binary metrics (see in Table 6), where the metrics are averaged across all the classes in many possible ways, such as micro, macro, etc. The former averaging method (micro) takes class imbalance into account, calculating the metrics globally by counting the number of times each class was correctly and incorrectly predicted. In contrast, the latter approach (macro) does not take imbalance into account, measuring each class's metrics independently and attaining their unweighted mean. Sometimes, the scholars employ k-fold cross-validation to verify the robustness of the trained model. In such a case, the obtained metrics from each fold are expressed as an average value with standard deviation [326], which can be expressed as Eq. (3). If the standard deviation has a higher value, it symbolizes higher inter-fold variation, consequently poor robustness and vice-versa.

$$M = \frac{1}{K} \times \sum_{i=1}^{K} M_i \pm \sqrt{\frac{\sum_{i=1}^{K} (M_i - \mu)^2}{K}}, \tag{3}$$

where $M_i \in \mathbb{R}$, $i \forall K$, denotes an estimated metric with a mean value of μ and K is the fold numbers. The further one-by-one exploration of the literature exposes that accuracy, sensitivity, and specificity are the most frequent metrics for evaluation. Those metrics are employed in 39.13%, 11.96%, and 14.13% literature, respectively. Although many articles use accuracy as a metric, it does not consider the class imbalance measurement. If there is a huge proportion of samples in one particular class, the ML models tend to learn that class more accurately, increasing the false-positive or false-negative rates. The conventional metrics dispense very high values if there is no penalization of class imbalance, which is accounted for in some metrics, for instance, ROC and AUC estimation [327]. However, barely 8.70% efforts consider this issue by determining the AUC values for their evaluation.

4. Discussion

This section gives an illustrative analysis of various reviewed methods from the above sections, with several shortcomings associated with scientific problems in the experimental scheme, which is considered future research objectives for the MVI methods (see in Section 4.1). Moreover, we explore the MVI method incorporations with the ML classifiers, aiming that MVI enhances the ML's results (see in Section 4.2). Additionally, we demonstrate enlightening recent research trends on

Table 6
List of the commonly-applied metrics for ML-based binary categorization models, collected from various literature.

Metrics	Mathematical formula and comments
TPR	$TRP = \frac{TP}{TP+FN} = 1 - FNR$, which measures the proportion of actual positives correctly identified as positives, without accounting for the co-relations between variables.
TNR	$TNP = \frac{TN}{TN+FP} = 1 - FPR$ holds the proportion of actual negatives that are correctly identified negatives, which also does not consider variables' co-relations.
FPR	$FPR = \frac{FP}{FP+TN} = 1 - TNR$, which is the probability of falsely rejecting the null hypothesis. It also estimates the Type-I error of a classifier.
FNR	$FNR = \frac{FN}{FN+TP} = 1 - TPR$ is the proportion of positives, which yield negative test outcomes with the test estimating the Type-II error of a classifier.
PPV	$PPV = \frac{TP}{TP+FP} = 1 - FDR$ is the proportions of positive observations, which are actually positives. It does not provide class-balanced estimations.
NPV	$NPV = \frac{TN}{TN+FN} = 1 - FOR$, which is the proportions of negative observations that are genuinely negative, and it varies with test sensitivity, test specificity, and disease prevalence.
FDR	$FDR = \frac{FP}{FP+TP} = 1 - PPV$ is the expected proportion of rejected null hypotheses that are incorrect rejections of the null, which controls the probability of Type-I error.
FOR	$FOR = \frac{FN}{FN+TN} = 1 - NPV$, which measures the proportion of the negatives results that are incorrectly rejected, controlling the probability of type-II error.
ACC	$ACC = \frac{TP+TN}{P+N}$ is a measure of how well a binary classification test correctly identifies or excludes a condition, which does not yield class-balanced estimations.
F1 Score	$F1 = 2 \cdot \frac{PPV \cdot TPR}{PPV+TPR} = \frac{2 \cdot TP}{2 \cdot TP+FP+FN}$, which is the harmonic mean of precision and recall, producing class-balanced estimations.
ROC	The ROC curve is plotted such that TPR is on the vertical axis and FPR is on the horizontal axis [325] by varying thresholds. It characterizes the tradeoff between hit rate and false alarm rate, designates the relationship between sensitivity and specificity.
Accuracy	$AUC = \sum_{i=1}^{N-1} \frac{1}{2} \cdot (FP_{i+1} - FP_i) \times (TP_{i+1} - TP_i)$ is the measure of the ability of a classifier to distinguish between classes, which is an unbiased class estimation and is a summary of the ROC curve.
PR	The PR curve is plotted such that PPV is on the vertical axis and TPR is on the horizontal axis by varying thresholds. It shows a tradeoff between precision and recalls for different thresholds, which applies in moderate to significant class imbalance cases.
AP	$AP = \sum_{i=1}^{N-1} (TPR_n - TPR_{n-1}) \times PPV_n$, which combines recall and precision for ranking, which describes the weighted mean of precision in each threshold with the increase in recall from the previous threshold used.

the discussed methods, accumulating the approximately past decade information from the collected 191 articles, and several recommendations for the further working areas (see in Section 4.3).

4.1. Analysis of MVI methods and their evaluation

Fig. 7 bestows the top twelve highly applied statistical and ML-based methods for the MVI, commonly employed in the literature from 2010 to August 2021. This figure reveals that EMMVI, MMVI, LLSMVI, BPCAMVI, and LRMVI are regularly selected as the top-5 statistical MVI techniques employed in 34, 34, 12, 11, and 11 articles, respectively. It is noteworthy that the statistical EMMVI and MMVI are most highly applied, as their utilization is approximately three times higher than the third MVI algorithm (LLSMVI). According to our study, those two methods should be deemed the representative baseline statistical MVI techniques, which was also the outcome of a survey from 2006 to 2017 by Lin and Tsai [2]. The possible reasons for those methods' utilization are: easy to implement and memory efficient, less affected by the outlier imputation as they hunt values within the attributes' ranges, needs negligibly less time for missing value prediction, does not require prior knowledge about the data, and imputed values keep the attributes' mean approximately unbiased. Although the EMMVI or MMVI imputation preserves the attribute's mean, they do not undertake the co-relation between the attributes, which may adversely affect the missingness imputation in some applications, especially where the attributes' mutual relations are crucial. Again, the MMVI is not a suitable MVI method for the categorical attribute values, as MMVI may return the fractional values, which necessitate being truncated with the lost information. On the contrary, the third and fifth most utilized, LLSMVI and LRMVI, estimate the associations among attributes, and then the regression coefficients are used to determine the missing values. Those two categories of MVI, such as LLSMVI and LRMVI, are practiced to predict numerical and categorical attribute values, as reported in many articles of the chosen 191 papers.

Furthermore, KNNMVI, RFMVI, KMCMVI, FCMMVI, and CARTMVI have commonly been used as top-5 ML-based MVI methods applied in 24, 12, 7, 7, and 7 articles, respectively. The KNNMVI is the most highly applied ML-based MVI process, as it is approximately two times higher than the second MVI algorithm (RFMVI), which can be marked as the typical representative baseline ML-based MVI method. The supervised techniques are KNNMVI, RFMVI, and CARTMVI, among the top five MVI methods, whereas the remaining KMCMVI and FCMMVI are unsupervised systems. The complete dataset is used as a training set in the former supervised systems, and the missing dataset is utilized as a testing set. In contrast, the latter unsupervised clustering methods group a set of related things into identical assemblages. Remarkably, the individual cluster center is the objects' mean in the same cluster. The distance between incomplete data and the recognized cluster centroids is determined to impute the missing values, where the nearest centroid's values are employed to fulfill in missing values. Such cluster methods are very time-consuming to reach the stopping criterion, requiring massive iterations if the initial guesstimation is too far from the real solutions. On the other hand, supervised KNNMVI has expeditious computation time, a simplistic, easy-to-understand algorithm, and evolving with new data, which could be the permissible causes for the massive utilization of the KNNMVI method. In conclusion, for the MVI methods, it is remarkable that statistical MVI techniques have been more employed in the last decade comparing the ML-based MVI systems, according to our survey. Again, Fig. 7 confirms that all top-12 statistical MVI systems are more employed than the corresponding ML-based MVI methods. The statistical MVI schemes do not require training in high configuration machines and a swift imputation process, which could be why they are preferred.

Evaluation is particularly crucial for validating the MVIs' performance and giving the ultimate conclusions. However, the summary of the MVI methods' evaluation metrics in Table 4 is displayed in Fig. 8, conferring the number of papers for direct and indirect evaluation against each year. This figure demonstrates that for all the years, except 2012, the direct evaluation methods are hugely applied through some metrics (see in Table 3). It is also noteworthy from Fig. 8 that after 2017 the direct evaluation systems are enormously utilized for the MVI process assessment. The proportion of relevant tasks that employ an indirect evaluation policy is hugely more negligible than the quantity of utilizing the direct judgment approach, as clearly depicted in Fig. 8. Additionally, the article numbers consider both evaluation procedures at the same time are indeed negligible, as also reviewed by Lin and

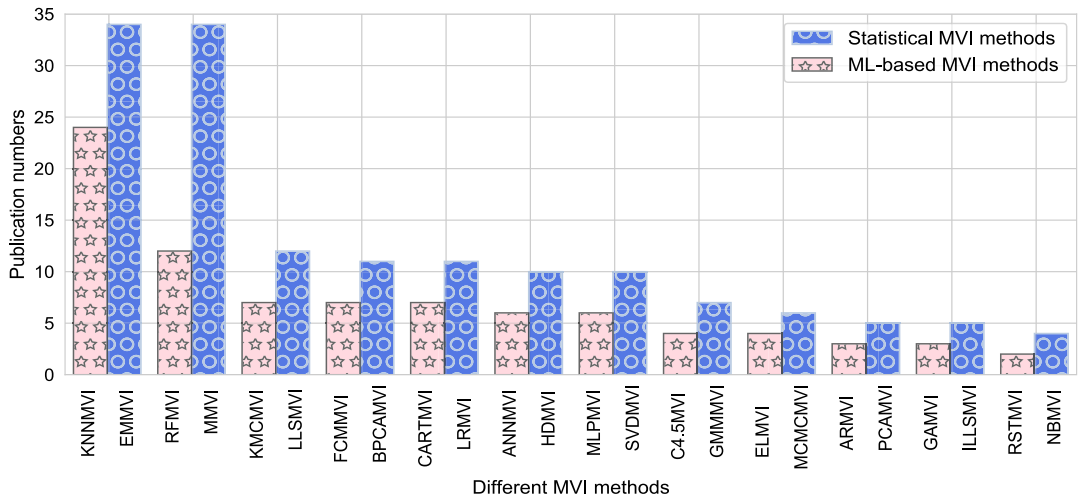


Fig. 7. Top twelve most commonly employed MVI methods in the order of highest to lowest utilizations. The highest to lowest statistical and ML-based methods are arranged side-by-side to compare those statistical and ML-based methods' appliances.

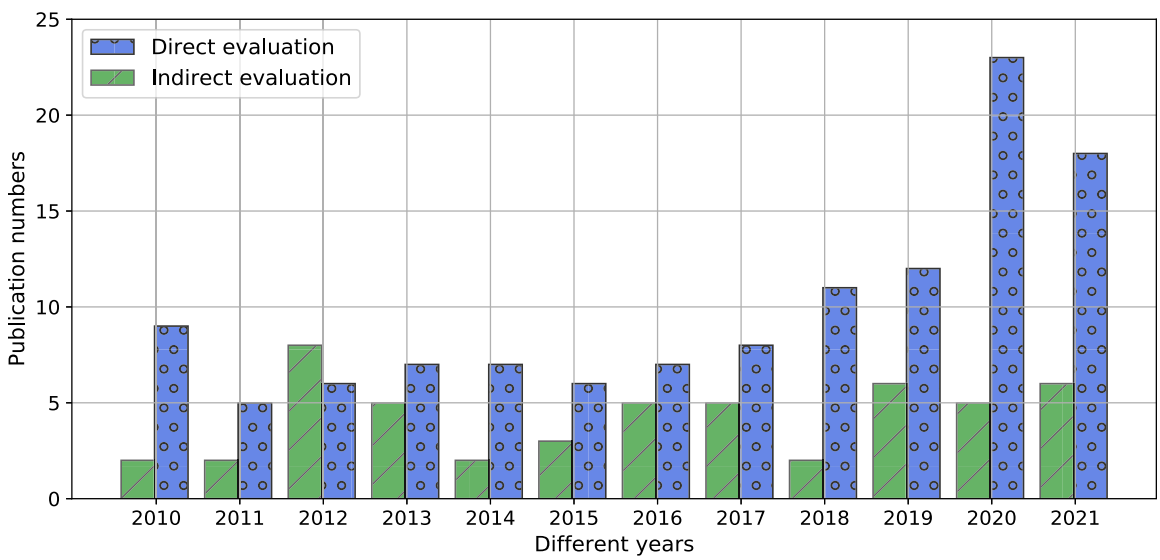


Fig. 8. Year-wise direct and indirect articles for evaluating MVI methods, where those two methods are presented side-by-side to compare their utilization for each year.

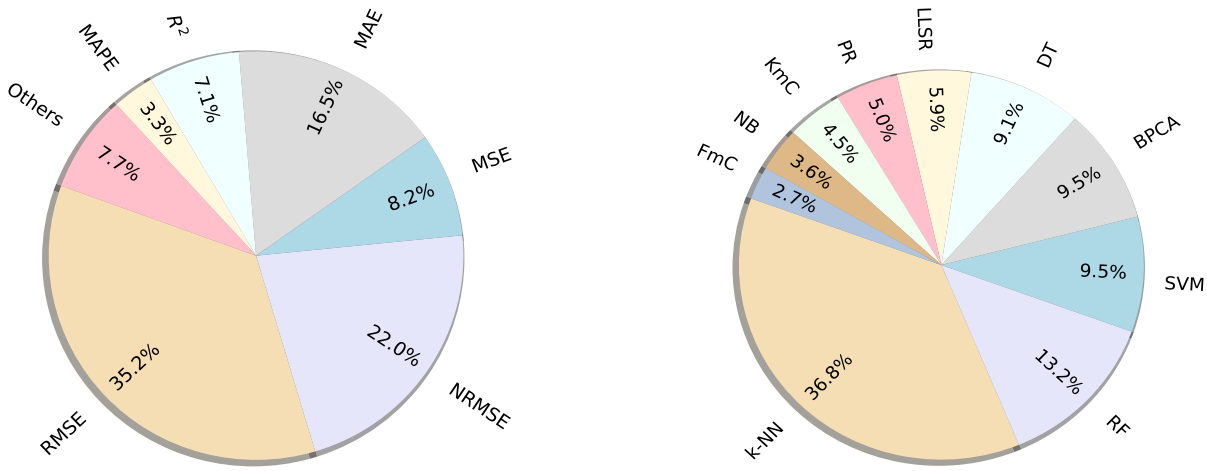
Tsai [2]. Some of the articles exercising both direct and indirect assessment policies are [86,95,180,186,188,197,199,209,211,228]. The further exploration of the elected paper tells that the most commonly employed direct metrics and classification models for the MVI policy assessment, which are displayed in the pie plot in Fig. 9. The left figure (Fig. 9(a)) indicates that the most usually exercised direct metrics are: RMSE, NRMSE, MSE, MAE, R^2, and MAPE. Only 7.7% articles practiced the rest of the metrics in Table 3 except those mentioned metrics in the last decade. It also points to be noted that the RMSE and NRMSE are alike metrics applied in the 57.2% articles. Therefore, according to the last decade's trends, the RMSE metric is a far better evaluation criterion for measuring the accuracy of the data missingness imputation. Again, Fig. 9(b) bestows the percentage of top-10 most widely employed ML models, such as KNN, RF, SVM, BPCA, DT, LLSR, PR, KmC, NB, and FCmC, for indirect evaluation. It is remarkable from Fig. 9(b) that KNN and RF models are the most widely employed ML models for indirect assessment in the last decade, encompassing a total of 50.0% articles from 2010 to August 2021 roughly.

4.2. Effect of MVIs on ML models

In Sections 4.2.1 and 4.2.2, we choose two different datasets for two different applications containing missing values and review their corresponding articles to determine the MVI's effect on their diagnostic or decision-making pipelines. Additionally, in Section 4.2.3, eleven diverse datasets of different areas are conferred, presenting the essence of MVI, which are accumulated from various previously published articles.

4.2.1. PIMA Indians Diabetes (PID) dataset

The PIMA Indians Diabetes (PID) dataset contains 768 female diabetic patients from the Pima Indian population near Phoenix, Arizona [328]. This PID dataset consists of 268 diabetic patients (positive) and 500 non-diabetic patients (negative) with eight attributes. Table 7 displays the statistical summary and description of this PID dataset. Out of the total eight attributes in Table 7, $F3$, $F4$, and $F5$ contain different number of missing values, such as 35, 227, and 374, respectively. In the following paragraph, we investigate five different articles to observe the performance improvements due to the appliance of MVIs.



(a) Metrics for direct MVI evaluation (b) ML models for indirect MVI assessment

Fig. 9. Pie plot of different metrics and ML models for evaluating MVI methods, dispensing their utilization percentage in the selected articles.

Table 7
The summary and a short description of the PID dataset proffering the attribute's mean and standard deviation (std).

SN	Attributes	Short description	Mean ± Std
1	Pregnant (F1)	Number of times pregnant	3.845 ± 3.369
2	Glucose (F2)	Plasma Glucose Concentration at 2 Hours in an Oral Glucose Tolerance Test	120.894 ± 31.973
3	Pressure (F3)	Diastolic Blood Pressure (mm-Hg)	69.105 ± 19.355
4	Triceps (F4)	Triceps Skin Fold Thickness (mm)	20.536 ± 15.952
5	Insulin (F5)	2-Hour Serum Insulin (μU/ml)	79.799 ± 115.244
6	BMI (F6)	Body Mass Index (Weight in kg/(Height in inches)^2)	32.00 ± 7.884
7	Pedigree (F7)	Diabetes Pedigree Function	0.472 ± 0.331
8	Age (F8)	Age of the patients (years)	33.241 ± 11.760

Hasan et al. [328] incorporated statistical mean imputation (i.e., MMVI) in their proposed pipeline. The authors trained different ML models, such as KNN, DT, RF, AdaBoost (AdB), NB, and XGB, with or without MVI. Their experimental results demonstrated that the incorporation of MVI improves the AUC values with a margin of 10.9% when the XGB classifier was employed. Wang et al. [329] combined supervised NB-based imputation in their recommended DMP_MI pipeline, which consists of oversampling and RF classifier. The authors confirmed that their imputation method improved the AUC value with a 5.8% margin using 5-fold cross-validation, comparing an RF classifier alone. A supervised KNN-based imputation (i.e., KNNMVI) was incorporated with the KNN classifier by Christobel and SivaPrakasam [330], dispensing 1.5% growth of accuracy due to the missing value imputation. The extensive experiments were conducted by Maniruzzaman et al. [126] for deciding different integral components, including different MVIs. The authors found that the RF classifier with group median imputation (i.e., MMVI) and others confers improved accuracy of 1.0%. Kandhasamy and Balamurali [331] jointed J48-DT classifier with mean value imputation (i.e., MMVI) and achieved 12.0% more accuracy for this aimed task. The summary of those papers for the ML model's performance improvements due to the MVI method employment is displayed in Fig. 10, which shows the efficacy of MVI method utilization.

4.2.2. Heart Disease (HD) dataset

The Heart Disease (HD) dataset is available in UCI Machine Learning Repository [332], in which the number of observations is 303 in two classes (sick with 164 samples and normal with 139 samples). Table 8 displays the statistical summary and description of all thirteen attributes of this HD dataset. Out of the total thirteen attributes in Table 8, $F7$, $F8$, $F10$ and $F11$ contain different number of missing values, such as 99, 147, 21, and 2, respectively. In the subsequent paragraph, we study five various articles on the HD dataset to scrutinize the performance enhancements due to the appliance of MVIs.

Nilashi et al. [332] merged two different MVI methods, such as ML-based KNNMVI and statistical HDMVI, with Fuzzy SVM classifier and other preprocessing, where the authors achieved 2.5% more accuracy. The KNNMVI also outperformed HDMVI with a margin of 1.1% in terms of accuracy in their recommended pipeline. Again, Khennou et al. [333] also employed ML-based KNNMVI method with SVM model, outputting better accuracy than without imputation by a margin of 11.6%. Setiawan et al. [334] applied ANNMVI with KNN classifier for this dataset, where they bestowed 5.6% rises in accuracy, due to the missing value imputation. A hybrid classifier with weighted voting was proposed by Saini et al. [335], where the authors imputed the missing values applying statistical MMVI. They experimentally showed that the integration of MMVI with their classifier magnified the accuracy with a boundary of 6.3%. Rani et al. [336] imputed the missing values using a statistical MICEMVI method and incorporated them with the AdB classifier and feature selection, where they reported improved accuracy for this dataset by the margin of 4.3%, corresponding to AdB alone. The review of those articles for the ML model's performance increases due to the MVI approach engaging is illustrated in Fig. 11, which reveals the effectiveness of MVI process employment.

4.2.3. Other datasets

We further investigate many articles for different domain datasets and methods in this section. The articles are collected in necessitating that the MVI methods are incorporated in the recommended pipeline for performance enhancement. The summary of this aimed task is manifested in Table 9. For all eleven different datasets, it is recognized that the employment of MVI systems with a good classifier increases the performance, where the authors employed diverse methods for MVI and classification. Again, those results reveal that the KNN and RF classifiers

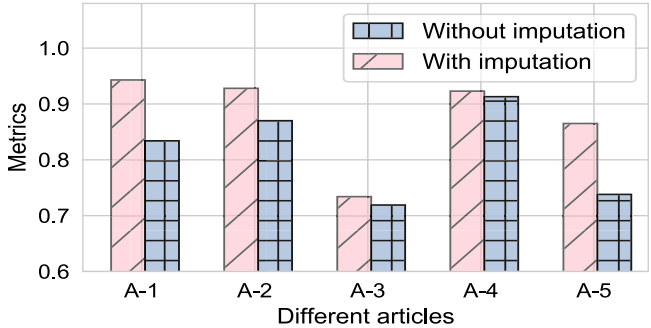


Fig. 10. Demonstration of ML's performance improvements due to MVI method incorporation for the PID dataset, where the articles A-1 to A-5, respectively, denote Hasan et al. [328], Wang et al. [329], Christobel and SivaPrakasam [330], Maniruzzaman et al. [126], and Kandhasamy and Balamurali [331].

Table 8
The summary and a short description of the HD dataset conferring the attribute's mean and standard deviation (std).

SN	Attributes	Short description	Mean ± Std
1	Age (F1)	Age of the patients in years	54.37 ± 9.08
2	Ca (F2)	Major vessel numbers (0–3) colored by fluoroscopy	0.73 ± 1.02
3	Chol (F3)	Serum cholesterol in mg/dl	246.26 ± 51.83
4	Cp (F4)	Chest pain type (4 values)	0.97 ± 1.03
5	Exang (F5)	Exercise induced angina	0.33 ± 0.47
6	Fbs (F6)	Fasting blood sugar >120 mg/dl	0.15 ± 0.36
7	Oldpeak (F7)	ST depression induced by exercise relative to rest	1.04 ± 1.16
8	RestECG (F8)	Resting electrocardiographic results with values of 0, 1, and 2	0.53 ± 0.53
9	Sex (F9)	Gender of the patients	0.68 ± 0.47
10	Slope (F10)	The slope of the peak exercise ST segment	1.40 ± 0.62
11	Thal (F11)	3, 6, and 7 mean normal, fixed defect, and reversible defect	2.31 ± 0.61
12	Thalach (F12)	Maximum heart rate achieved	149.65 ± 22.91
13	Trestbps (F13)	Resting blood pressure in mm-Hg on admission to hospital	131.62 ± 17.54

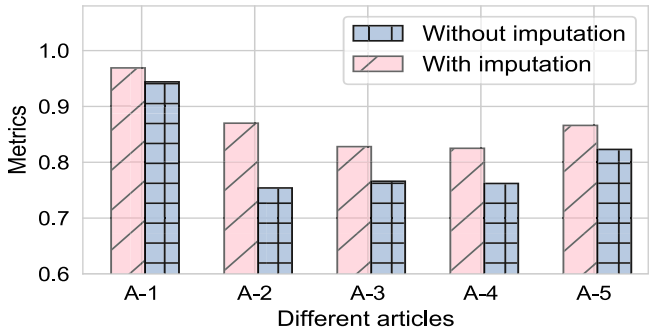


Fig. 11. Demonstration of ML's performance improvements due to MVI method incorporation for HD dataset, where the articles A-1 to A-5, respectively, denote Nilashi et al. [332], Khennou et al. [333], Setiawan et al. [334], Saini et al. [335], and Rani et al. [336].

Table 9
Classification results from different articles with/without MVI methods, where various MVI systems and ML models were employed. The upward arrow (↑) indicates that the higher value is a better outcome, and the downward arrow (↓) is vice-versa.

Articles	Dataset types	MVI methods	Classifiers	Results (MVI?)	
				No	Yes
EL-Bohy et al. [337]	Hepatitis	MMVI	NB	0.832 ↑	0.888 ↑
Ribeiro and Freitas [229]	Stroke	ANNMVI	RF	0.603 ↑	0.713 ↑
Ribeiro and Freitas [229]	Heart Attack	ANNMVI	RF	0.646 ↑	0.718 ↑
Ribeiro and Freitas [229]	Parkinsons	ANNMVI	RF	0.601 ↑	0.677 ↑
Zeng et al. [186]	TCM medical	KNNMVI	RF	0.742 ↑	0.803 ↑
Darryl and Rahman [252]	Cardiovascular	KNNMVI	KNN	0.770 ↑	0.800 ↑
Pan et al. [145]	Tic-Tac-Toe	KNNMVI	NB	0.717 ↑	0.747 ↑
Kumari and Singh [338]	Breast cancer	MMVI	KNN	0.986 ↑	0.993 ↑
Minakshi et al. [35]	Unspecified	KNNMVI	C4.5	0.320 ↓	0.301 ↓
Tsai and Chang [162]	Unspecified	KNNMVI	KNN	0.588 ↑	0.622 ↑
Hasan et al. [172]	Measles	MMVI	LGBM	0.738 ↑	0.779 ↑

are primarily employed, which is the similar outcomes of Fig. 6. It is also remarked that the statistical MVI methods are widely applied for the missingness imputation, which essentially enhances the prediction results.

4.3. Observations and recommendations

This section reveals our critical observations, obtained from point-to-point scrutinization of the selected 191 articles thoroughly. Consequently, we point out several recommendations for the imputation of

missing value(s), which will serve as future guidance for beginner to expert-level scholars to build their decision-making framework from the incomplete datasets. The notable observations and recommendations are as follows:

- The necessity of the MVI process depends on how vital the missed value(s) are. There are several novel classification or regression approaches, which do not need the MVI method to handle the incomplete datasets [156,339], as they can tackle data missingness by themselves internally. At the beginning of pipeline development, the researcher can obtain approximate classification results from those models, utilizing the incomplete and complete datasets (imputed by this article's suggested representative baseline MVI method(s)). Such an experiment can indicate the importance of the missed value(s) and consequently provide the necessity of the missingness imputation or not?
- From the review analysis of this article, it may be remarked that, unfortunately, there is no specific clarification to the question: which method is the best for the missingness imputation? The scholars have to search for the most suitable method, as the MVI methods depend on various constituents of interest. However, the investigation of the selected 191 articles and another 21 articles (in Section 4.2), where the authors incorporated different MVI techniques to enhance the ML models' performance, explicate that the adoption of MVI methods depends on the many factors or other methods: attribute selection, outlier detection, attribute normalization, the classifier(s), the field of application(s), computational resource availability, time to receive imputation, and objective factor consideration (either mean conserving or inter-attributes' correlation preserving).
- Employment of attribute and/or sample selection before or after the MVI process could affect the imputation results, as it aims to remove unrepresentative attributes and/or samples from the provided feature vector. Doing one or both of these jobs before the MVI process could produce the cleaner complete dataset for the learning phase, probably outputting more trustworthy imputation returns. Alternatively, operating attribute and/or sample selection over an imputed dataset after MVI could make the classifier more beneficial than one based on the imputed dataset without attribute and/or sample determination. However, there are some articles considering those effects, which are [121,162,212,340]. Huang et al. [121],Tsai and Chang [162] explicated that MVI after the sample selection produced better classification results than MVI alone. Huang et al. [121],Liu et al. [212] revealed that attribute selection followed by the MVI process does not necessarily have a convincing impact while the MVI process followed by attribute selection has a positive influence. Lastly, Tran et al. [340] experimentally proved that the MVI method followed by attribute and sample selection improved the accuracy of the classification and massively reduced computational time.
- The data normalization or standardization can profoundly impact the imputation process, which seldom depends on the imputation technique. For instance, if distance-based methods (e.g., KNN-MVI) are applied, it is recommended to standardize the data and then impute it; attributes' lower values converge faster, leading to less computational time in real-time application. There is a controversy that should we impute first or standardize first? If standardization is applied first, the imputation process could influence the actual center and scale to use. On the other hand, the employment of imputation first can reduce the skewness of the estimated mean and scale due to missingness in the pattern.
- The imputed values are sometimes unreliable due to the presence of outliers, which has an adverse effect on the values entered for missing data. Therefore, outlier handling must be conducted before missingness imputation. Remarkably, the regression-type MVI methods are heavily affected by the outliers, demanding mandatory outlier rejection. On the other hand, the median value-based MVI methods can mitigate outliers' influences, as it imputes the missing values from the most frequent values, which may not be outliers.
- The direct assessment system, the specified metric(s) of the classifiers in indirect evaluation, and reflection of TCT are the three most common procedures to judge the MVI method(s). In order to thoroughly explain the performance of the MVI technique(s) and provide suggestions for developing more reliable imputation ways, all three of these evaluation systems should be applied. Unfortunately, it is rarely applied in related studies for the assessment in the last decade. This is one of the shortcomings of current practice in the articles, and the researcher should consider them together in future experimentation.

5. Conclusion

The MVI method for incomplete datasets is a significant problem in data mining, big data analysis, and ML-based decision-making pipeline, as the ultimate mining or analysis result could be adversely influenced if the incomplete datasets are not suitably imputed. This article aims to review and investigate 191 related articles published from 2010 to August 2021. The review and analysis of this article focus on the issues confronted during the MVI process, such as the MVI procedures applied and the evaluation schemes contemplated. The investigation of the selected last decade's articles exhibits many challenges in the existing literature, such as which methods are best for MVI and its evaluation, what factors can unfavorably affect the missingness imputation, and how the MVI methods influence the decision-making pipeline. The investigated conclusions from the selected 191 MVI's articles reveals that EMMVI, HDMVI, LLSMVI, LRMVI, MMVI, MICEMVI, BPCAMVI, SVD-MVI, ANNMVI, KMCMVI, FCMMVI, CARTMVI, KNNMVI, and RFMVI, are the most regularly practiced MVI strategies in the last decade (2010–August 2021). The results also expose that the statistical MVIs like EMMVI, MMVI, LLSMVI, BPCAMVI, and LRMVI are more employed methods, as they do not require training in high configuration machines and a swift imputation process. Moreover, for the direct MVI evaluation, RMSE, NRMSE, MSE, MAE, and R^2 are the top five-most metrics. In contrast, KNN, RF, SVM, BPCA, and DT are the top five-most ML models employed for indirect MVI evaluation. Expectedly, the outcomes of this article will be helpful to the related research community for recovering those challenges, selecting a suitable MVI method, and its evaluation metric. The recommendations may also be an excellent command for future researchers to develop an effective decision-making system using ML model(s) and incomplete datasets for real-life applications in many domains.

CRediT authorship contribution statement

Md. Kamrul Hasan: Conceptualization, Methodology, Formal analysis, Investigation, Writing – review & editing, Visualization, Supervision. **Md. Ashraful Alam:** Resources, Writing – original draft. **Shidhartho Roy:** Resources, Visualization, Writing – original draft. **Aishwariya Dutta:** Resources, Writing – original draft. **Md. Tasnim Jawad:** Resources, Writing – original draft. **Sunanda Das:** Resources, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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